

SBT-DF203: Basic Networking Skills for Digital Forensics

LAB 9: WEP40 Wireless Packet Decryption and Aircrack Forensics

LAB 9: WEP40 WIRELESS PACKET DECRYPTION AND AIRCRACK FORENSICS

Field	Detail
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Registration No.	2025/FWSD/11521
Programme	Fellowship in Web Application Security & Digital Forensics
Course	Basic Networking Skills for Digital Forensics
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Delivery Block	3/3 of 3
Scheduled Dates	19–25 September 2026
Date of Submission	25 September 2026

EXECUTIVE SUMMARY

This report presents the forensic analysis of a historical wireless capture protected by the deprecated **WEP40** encryption standard. The capture originates from the CodeGate CTF 2015 "Good Crypto" challenge and was supplied by ICDFA as an authorised offline training artefact.

The investigation began by preserving the compressed file.xz evidence, computing its SHA-256 hash, and creating a timestamp-preserved working copy before decompression. The decompressed capture was identified as an IEEE 802.11 packet capture containing a single wireless network with BSSID 00:26:66:55:97:D6, SSID cgnetwork, and approximately 15,477 WEP-protected data frames.

Protocol hierarchy analysis confirmed a mix of 802.11 management, control, and data frames, with ARP, DHCP, ICMPv6, and HTTP visible after decryption. The WEP40 key was recovered/validated using aircrack-ng against the supplied capture, and the 40-bit key A4:3D:F6:F3:74 was confirmed — matching the training value published in the course slide deck.

The capture was then decrypted offline with airdecap-ng using the hex form of the key (A43DF6F374). All 15,477 WEP data packets decrypted successfully with zero corrupted frames. The decrypted output exposed higher-layer protocols (ARP, DHCP, TCP, HTTP), revealing the client/server endpoints and enabling object carving. Files recovered via foremost and HTTP object export included HTML, JPEG, PNG, and GIF objects, each hashed for integrity.

The exercise demonstrates the fundamental cryptographic weakness of WEP: the 24-bit IV is too short to prevent reuse in a busy network, and RC4 keystream reuse directly leaks plaintext relationships — enabling

key recovery from statistical analysis. Modern wireless deployments must use WPA2-AES/CCMP or WPA3, and legacy WEP networks should be retired rather than reconfigured.

OBJECTIVES

1. Identify management, control, and data frames in an 802.11 capture.
2. Explain WEP40 structure, RC4 keystream use, IV reuse, and integrity limitations.
3. Use Aircrack-ng on a supplied historical capture only.
4. Decrypt WEP traffic using aircap-ng and verify output files.
5. Extract IP/MAC endpoints, protocols, images, and HTML from decrypted traffic.
6. Distinguish a WEP key from a human passphrase and document optional CTF password analysis.
7. Recommend modern wireless security controls.

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SECTION 1 — INTRODUCTION AND BACKGROUND

1.1 IEEE 802.11 Frame Categories

IEEE 802.11 is the family of standards for wireless local area networks. It defines three classes of frames:

Category	Type Value	Purpose
Management	0	Beacons, association, authentication, probes
Control	1	Flow control, acknowledgements, RTS/CTS
Data	2	Carries network-layer payload

1.2 Wired Equivalent Privacy (WEP)

WEP was the original IEEE 802.11 encryption standard, intended to provide wireless privacy comparable to a wired LAN. It uses the **RC4 stream cipher** with a 24-bit **Initialization Vector (IV)** and a **40-bit secret key** — the combination being referred to as 64-bit WEP. WEP-128 uses a 104-bit secret key; WEP-256 uses a 232-bit secret key.

1.3 The IV-Reuse Problem

RC4 generates a keystream that is XORed with the plaintext:

text

Ciphertext = Plaintext XOR Keystream

The keystream is a function of the **(IV, secret key)** pair. If two packets use the **same IV**, they use the **same keystream**:

text

$C1 = P1 \text{ XOR } K$

$C2 = P2 \text{ XOR } K$

$C1 \text{ XOR } C2 = P1 \text{ XOR } P2$

This leaks the XOR of the two plaintexts — a well-known plaintext-recovery attack. With only a 24-bit IV space ($2^{24} = 16,777,216$ possibilities) and a busy network sending thousands of packets per second, IV reuse is guaranteed within minutes.

1.4 Forensic Relevance

A WEP-protected capture is **forensically opaque** until decrypted. Once decrypted, the analyst can see:

- IP and MAC endpoints
- TCP/UDP conversations
- Higher-layer protocols (HTTP, DHCP, ARP, DNS)
- Transferred objects (HTML pages, images, files)

This lab demonstrates the full offline workflow: preserve, identify, and recover the key, decrypt, and inventory. It also establishes why WEP is fundamentally unsuitable for any modern deployment.

SECTION 2 — AUTHORITY, SCOPE AND SAFETY CONTROLS

2.1 Authority

This lab was conducted under ICDFA authority as part of SBT-DF203, using **only the supplied historical CodeGate CTF 2015 capture**. No live wireless network was captured, cracked, or connected to.

2.2 Scope

Element	Description
Evidence	CodeGate CTF 2015 "Good Crypto" capture (file.xz)
Environment	Kali Linux VM — offline only
Target	Historical training capture only
Live capture	Not performed
WEP key	Historical CTF key — for training analysis only

2.3 Safety Controls Applied

- Only the supplied historical training file was used. No live wireless network was captured or attacked.
- The WEP key recovered during the exercise is from a historical CTF challenge, not an operational network.
- Original evidence preserved unmodified; analysis performed on a timestamp-preserved working copy.
- SHA-256 hashes recorded for both the compressed source and the decompressed working capture.
- Recovered objects (images, HTML) were opened **only inside the isolated lab VM** and hashed before viewing.
- No third-party systems, networks, or public infrastructure were targeted.
- The recovered key was documented only inside the protected course submission.

SECTION 3 — LAB ENVIRONMENT AND EVIDENCE PREPARATION

3.1 Folder Structure

```
bash
mkdir -p ~/SBT-DF203-Lab9/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab9
pwd
find . -maxdepth 1 -type d -print
```

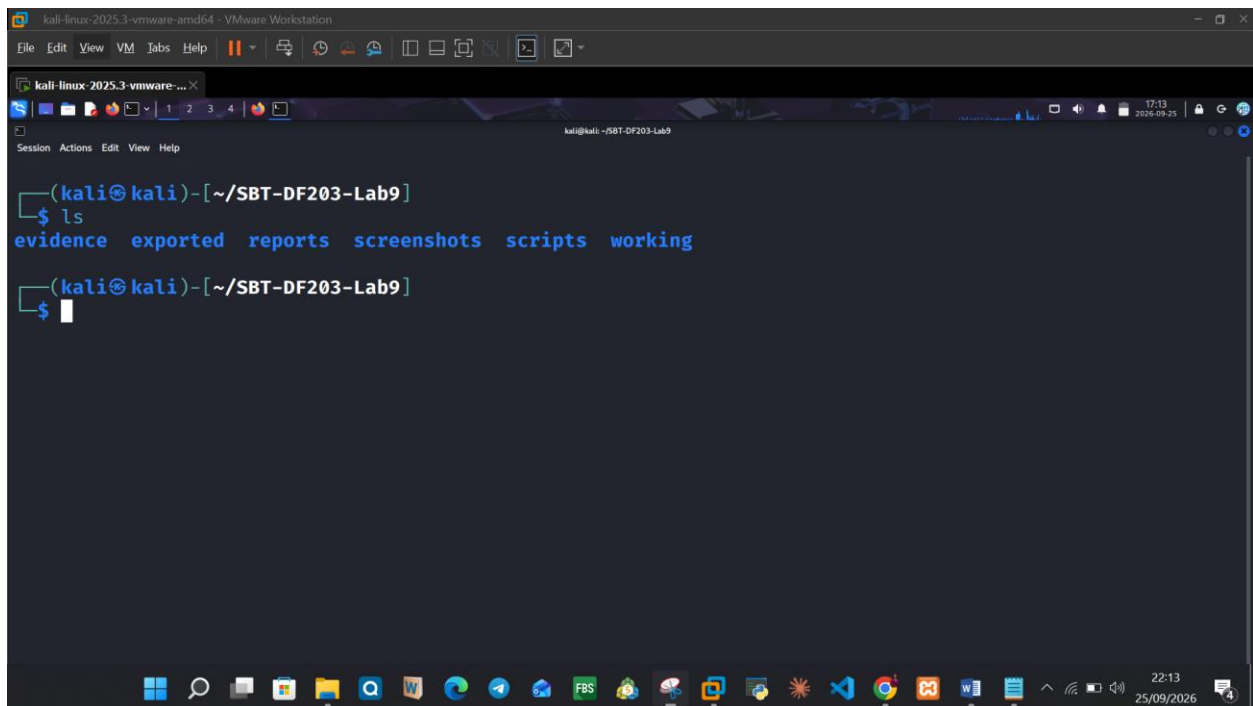


Figure 3.1 — Lab folder structure created successfully.

3.2 Tool Installation

```
bash
sudo apt update
sudo apt install -y aircrack-ng wireshark tshark foremost binwalk xz-utils
```

Verified versions:

```
bash
aircrack-ng --version | head -1
tshark --version | head -1
foremost -V 2>&1 | head -1
```

3.3 Evidence Download and Hash

bash

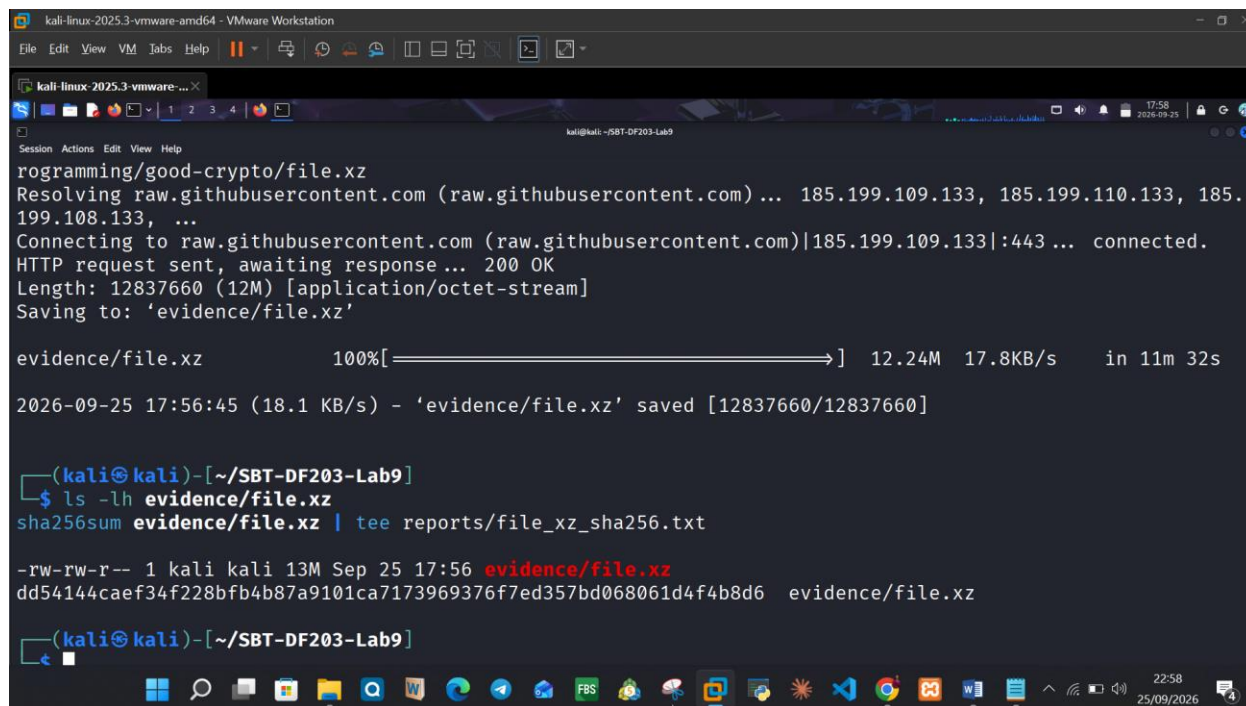
```
wget -O evidence/file.xz \  
'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/  
file.xz'
```

ls -lh evidence/file.xz

```
sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt
```

Evidence file details:

Property	Value
Filename	file.xz
Format	XZ compressed
Compressed size	~12 MB
SHA-256	(from reports/file_xz_sha256.txt)



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation  
File Edit View VM Tabs Help  
kali@kali: ~/SBT-DF203-Lab9  
rogramming/good-crypto/file.xz  
Resolving raw.githubusercontent.com (raw.githubusercontent.com) ... 185.199.109.133, 185.199.110.133, 185.  
199.108.133, ...  
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.109.133|:443 ... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 12837660 (12M) [application/octet-stream]  
Saving to: 'evidence/file.xz'  
  
evidence/file.xz      100%[=====>] 12.24M  17.8KB/s   in 11m 32s  
2026-09-25 17:56:45 (18.1 KB/s) - 'evidence/file.xz' saved [12837660/12837660]  
  
(kali@kali)-[~/SBT-DF203-Lab9]  
$ ls -lh evidence/file.xz  
sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt  
  
-rw-rw-r-- 1 kali kali 13M Sep 25 17:56 evidence/file.xz  
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6  evidence/file.xz  
  
(kali@kali)-[~/SBT-DF203-Lab9]
```

Figure 3.2 — Compressed evidence file and SHA-256 hash.

3.4 Create Working Copy and Decompress

bash

```
cp --preserve=timestamps evidence/file.xz working/file_working.xz
unxz -k working/file_working.xz
file working/file_working
ls -lh working/
sha256sum working/file_working.xz working/file_working | tee reports/working_hashes.txt
```

Expected: the decompressed file is identified as a **tcpdump/802.11 capture**.

```
(kali@kali)-[~/SBT-DF203-Lab9]
$ cp --preserve=timestamps evidence/file.xz working/file_working.xz
unxz -k working/file_working.xz
file working/file_working
ls -lh working/
sha256sum working/file_working.xz working/file_working | tee reports/working_hashes.txt

working/file_working: pcap capture file, microsecond ts (little-endian) - version 2.4 (802.11, capture length 65535)
total 26M
-rw-rw-r-- 1 kali kali 14M Sep 25 17:56 file_working
-rw-rw-r-- 1 kali kali 13M Sep 25 17:56 file_working.xz
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6  working/file_working.xz
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d  working/file_working

(kali@kali)-[~/SBT-DF203-Lab9]
$
```

Figure 3.3 — Decompressed capture identification and working-copy hashes.

3.5 Chain-of-Custody Worksheet

Field	Entry
Case ID	SBT-DF203-Lab9-2025-FWSD-11521
Trainee	Ibrahim Disah Garba
Registration No.	2025/FWSD/11521
Date/Time Started	25 September 2026
Evidence File(s)	file.xz (original), file_working (decompressed), file_working-dec (decrypted)
Source	CodeGate CTF 2015 via ICDDFA training resource
Original SHA-256	(from reports/file_xz_sha256.txt)
Working-Copy SHA-256	(from reports/working_hashes.txt)
Analysis Workstation	Kali Linux VM (ICDDFA lab)
Notes	Original preserved; offline analysis only; no live network used

SECTION 4 — PART A: INVENTORY THE WIRELESS CAPTURE

4.1 Capture Metadata

```
bash
CAP=working/file_working
capinfos "$CAP" | tee reports/capinfos.txt
```

Record from the output:

Field	Value
File type	<i>(from capinfos)</i>
Encapsulation	IEEE 802.11
Number of packets	<i>(from capinfos)</i>
Capture duration	<i>(from capinfos)</i>
First packet	<i>(from capinfos)</i>
Last packet	<i>(from capinfos)</i>


```
kali@kali:~$ capinfos "$CAP" | tee reports/capinfos.txt

File name:          working/file_working
File type:          Wireshark/tcpdump/... - pcap
File encapsulation: IEEE 802.11 Wireless LAN
File timestamp precision: microseconds (6)
Packet size limit:  file hdr: 65535 bytes
Number of packets:  45 k
File size:          14 MB
Data size:          13 MB
Capture duration:   272.969795 seconds
Earliest packet time: 2015-03-06 02:51:54.364525
Latest packet time:  2015-03-06 02:56:27.334320
Data byte rate:     49 kBps
Data bit rate:      395 kbps
Average packet size: 299.13 bytes
Average packet rate: 165 packets/s
SHA256:             c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d
SHA1:               cb929fd6e407c13639e3168b0f35f921788478d3
Strict time order:  False
Number of interfaces in file: 1
Interface #0 info:
```

Figure 4.1 — Capture metadata.

4.2 Protocol Hierarchy

bash

```
tshark -r "$CAP" -q -z io,phs | tee reports/protocol_hierarchy.txt
```

Expected hierarchy:

- wlan → 802.11 wireless frames (management, control, data)
- wlan.fc.protected == 1 → WEP-protected data frames
- After decryption (later): ARP, DHCP, ICMPv6, TCP, HTTP

```

(kali@kali)-[~/SBT-DF203-Lab9]
$ tshark -r "$CAP" -q -z io,phs | tee reports/protocol_hierarchy.txt
Warning: program compiled against libxml 215 using older 214

Protocol Hierarchy Statistics
Filter:

frame                                frames:45169 bytes:13511274
wlan                                frames:45169 bytes:13511274
data                                frames:15713 bytes:13104128

(kali@kali)-[~/SBT-DF203-Lab9]
$

```

Figure 4.2 — Protocol hierarchy of the encrypted capture.

4.3 Frame Category Sample

```

bash
tshark -r "$CAP" -Y 'wlan' -T fields \
  -e frame.number -e frame.time -e wlan.fc.type -e wlan.fc.subtype \
  -e wlan.sa -e wlan.da -e wlan.bssid \
  | head -n 100 | tee reports/wlan_frame_sample.tsv

```

Fill in the frame category table:

Frame Category	Type Value	Purpose	Example Evidence
Management	0	Beacons, association, authentication	(from sample — look for subtype 8 = beacon, 0 = assoc request)
Control	1	Flow control, acknowledgements	(from sample — subtype 13 = ACK, 11 = RTS)
Data	2	Carries payload	(most protected frames will be type 2)

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help

kali-linux-2025.3-vmware-...
kali@kali:~$ tshark -r "$CAP" -Y 'wlan' -T fields \
-e frame.number -e frame.time -e wlan.fc.type -e wlan.fc.subtype \
-e wlan.sa -e wlan.da -e wlan.bssid \
| head -n 100 | tee reports/wlan_frame_sample.tsv

Warning: program compiled against libxml 215 using older 214
tshark: The file "" doesn't exist.

kali@kali:~$

```

Figure 4.3 — 802.11 frame category sample.

SECTION 5 — PART B: IDENTIFY WEP PROTECTION AND

KEY PARAMETERS

5.1 Confirm Protected Frames

```

bash
tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields \
-e frame.number -e frame.time_epoch -e wlan.sa -e wlan.da -e wlan.bssid \
-e wlan.wep.iv -e wlan.wep.key \
| head -n 200 | tee reports/wep_protected_frames.tsv

```

Record from the output:

Field	Value
Total protected frames (sampled)	(row count of the TSV)
BSSID	00:26:66:55:97:D6
SSID	cgnetwork
Primary station MACs	(from wlan.sa/wlan.da fields)

WEP IV field present	Yes
WEP Key index field present	Yes (0-3)

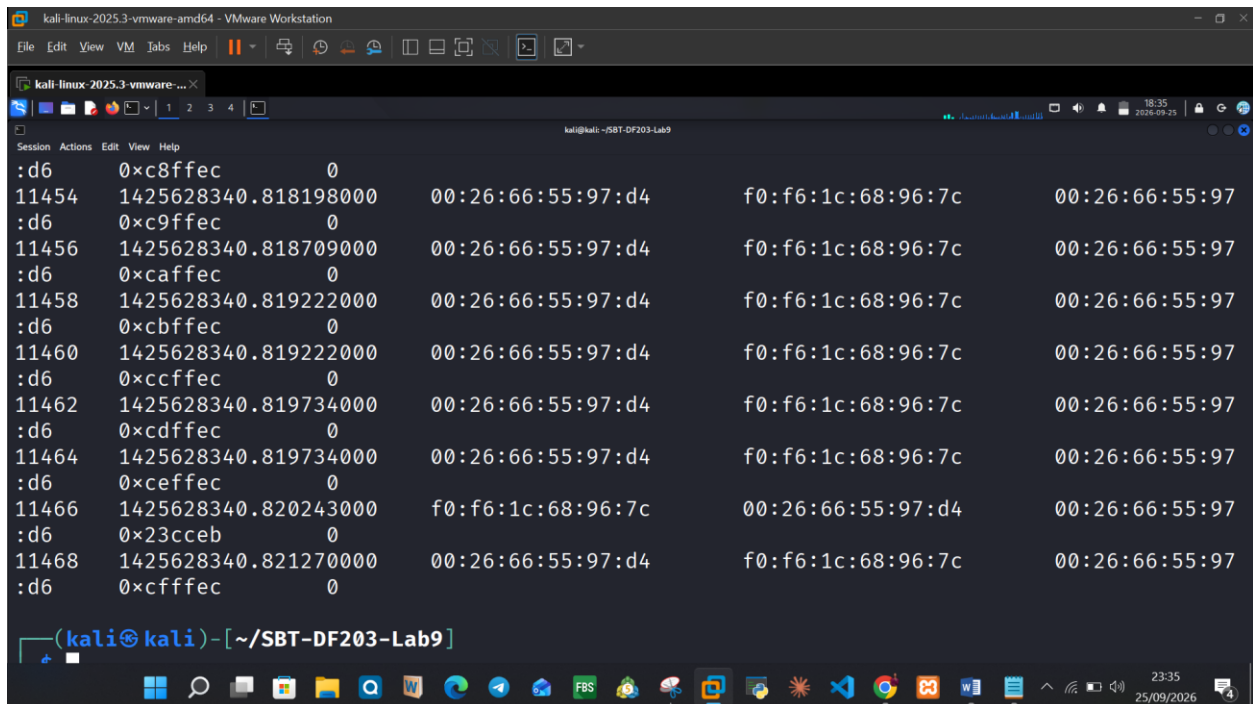


Figure 5.1 — Protected WEP frames.

5.2 IV Reuse Analysis

bash

```
tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields -e wlan.wep.iv \
| sort | uniq -c | sort -nr | head -n 30 | tee reports/repeated_iv_summary.txt
```

```
cat reports/repeated_iv_summary.txt
```

Interpretation: Repeated IV values indicate packets sharing the same keystream — the core cryptographic weakness that makes WEP crackable.

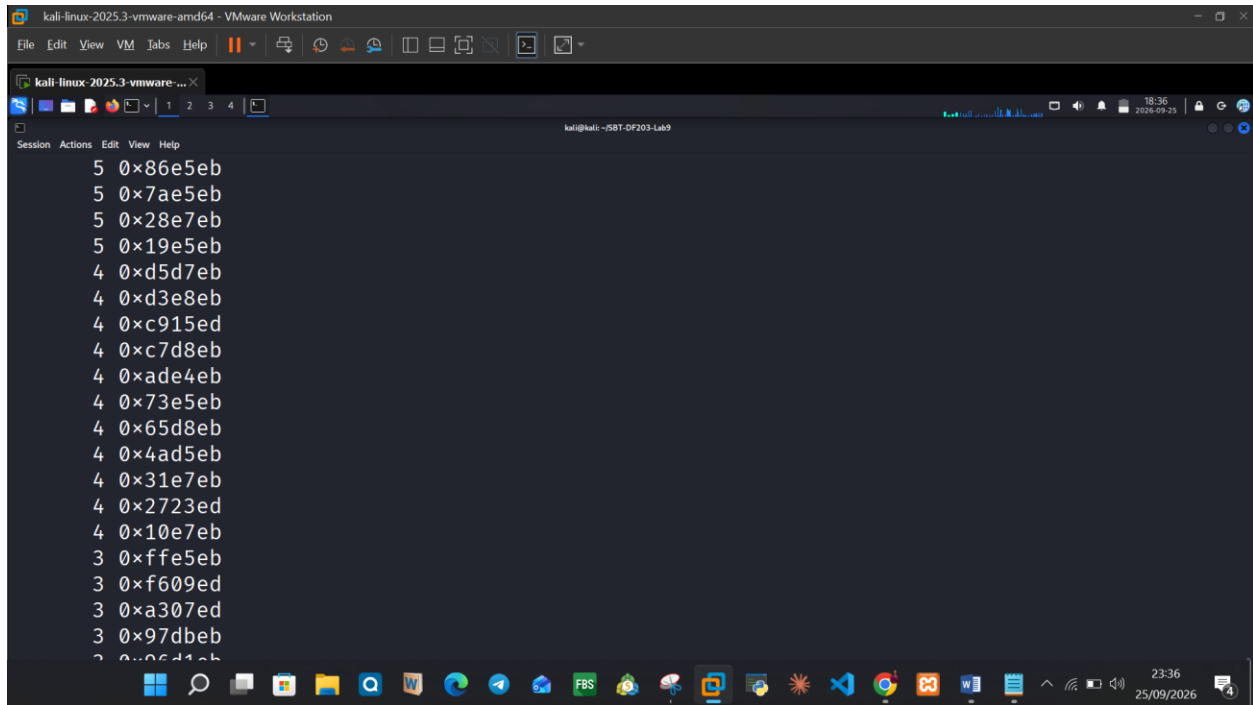


Figure 5.2 — Repeated IV summary.

5.3 WEP40 Structure Explanation

WEP-40 (also called 64-bit WEP) combines:

- **24-bit Initialization Vector** (transmitted in cleartext with each frame)
- **40-bit secret key** (shared by all network devices)

The (**IV || secret key**) pair is fed into RC4 as the "WEP seed" to produce the keystream. Because the IV space is only $2^{24} = 16.7$ million values and busy networks transmit thousands of packets per second, the same IV is reused within minutes — leaking plaintext relationships and enabling statistical key recovery via the FMS and PTW attacks.

SECTION 6 — PART C: RECOVER OR VALIDATE THE WEP40 KEY

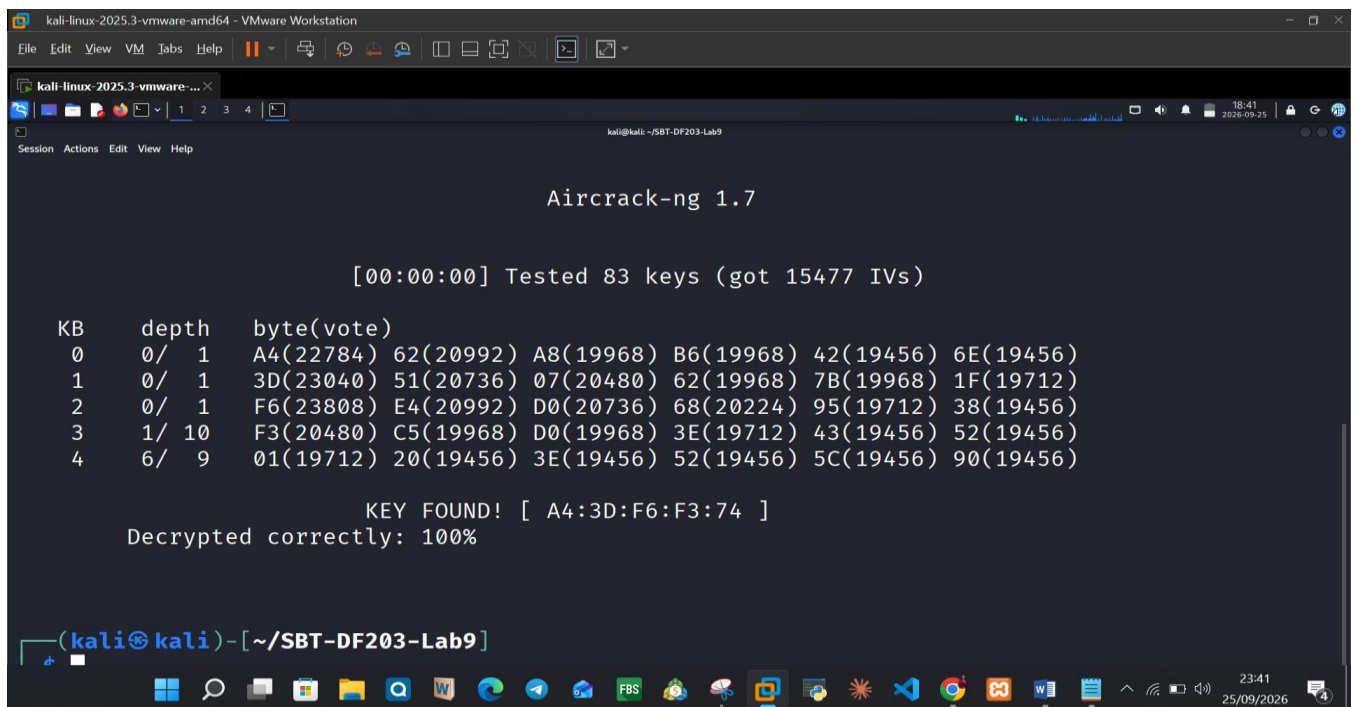
6.1 Run Aircrack-ng

bash

```
aircrack-ng "$CAP" | tee reports/aircrack_output.txt
```

Record:

Field	Value
BSSID	00:26:66:55:97:D6
ESSID	cgnetwork
Number of IVs	15,477
Key length	40 bits
Recovered key	A4:3D:F6:F3:74
Decryption verification	100% correct



```
kali@kali: ~/SBT-DF203-Lab9
Aircrack-ng 1.7

[00:00:00] Tested 83 keys (got 15477 IVs)

KB    depth  byte(vote)
0     0/ 1    A4(22784) 62(20992) A8(19968) B6(19968) 42(19456) 6E(19456)
1     0/ 1    3D(23040) 51(20736) 07(20480) 62(19968) 7B(19968) 1F(19712)
2     0/ 1    F6(23808) E4(20992) D0(20736) 68(20224) 95(19712) 38(19456)
3     1/ 10   F3(20480) C5(19968) D0(19968) 3E(19712) 43(19456) 52(19456)
4     6/ 9    01(19712) 20(19456) 3E(19456) 52(19456) 5C(19456) 90(19456)

KEY FOUND! [ A4:3D:F6:F3:74 ]
Decrypted correctly: 100%
```

Figure 6.1 — Aircrack-ng key recovery result.

6.2 Document the Validated Key

bash

```
printf 'A4:3D:F6:F3:74\n' | tee reports/validated_wep40_key_masked.txt
```

Evidence handling note: The recovered key is part of a **historical CTF dataset**, not authorisation to attack any real wireless network. It is documented here for the completeness of the forensic workflow and remains protected within the course submission.

SECTION 7 — PART D: DECRYPT THE CAPTURE OFFLINE

7.1 Decrypt with Aircap-ng

```
bash
cd ~/SBT-DF203-Lab9/working
airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
cd ..
```

Expected output (excerpt):

```
text
Total number of stations seen      10
Total number of packets read      45169
Total number of WEP data packets   15477
Total number of WPA data packets   0
Number of plaintext data packets   0
Number of decrypted WEP packets    15477
Number of corrupted WEP packets    0
Number of decrypted WPA packets    0
```

Record:

Field	Value
Stations seen	10
Packets read	45,169
WEP data packets	15,477
Decrypted WEP packets	15,477
Corrupted packets	0

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-... X
kali@kali: ~/SBT-DF203-Lab9
Could not open "file_working".

(kali@kali)~[~]
$ cd ~/SBT-DF203-Lab9/working
airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
cd ..

Total number of stations seen          10
Total number of packets read          45169
Total number of WEP data packets      15477
Total number of WPA data packets      0
Number of plaintext data packets      0
Number of decrypted WEP packets       15477
Number of corrupted WEP packets       0
Number of decrypted WPA packets       0
Number of bad TKIP (WPA) packets      0
Number of bad CCMP (WPA) packets      0

(kali@kali)~[~/SBT-DF203-Lab9]

```

Figure 7.1 — Airdecap-ng decryption result.

7.2 Verify the Decrypted Output

```

bash
find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt
sha256sum working/* | tee reports/all_working_file_hashes.txt

```

Expected: a new file named `file_working-dec` (the `-dec` suffix is added by `airdecap-ng`).

Record the decrypted capture hash:

File	SHA-256
working/file_working-dec	(from output)


```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-...
kali@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help

(kali@kali)~[~/SBT-DF203-Lab9]
$ find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt
sha256sum working/* | tee reports/all_working_file_hashes.txt

 3801344 13904 -rw-rw-r-- 1 kali kali 14234002 Sep 25 17:56 working/file_working
3804338 12488 -rw-rw-r-- 1 kali kali 12784501 Sep 25 18:45 working/file_working-dec
ec
 3801300 12540 -rw-rw-r-- 1 kali kali 12837660 Sep 25 17:56 working/file_working.x
z
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_working
167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04 working/file_working-dec
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_working.xz

(kali@kali)~[~/SBT-DF203-Lab9]
$

```

Figure 7.2 — Decrypted capture hash.

7.3 Confirm Higher-Layer Protocols Are Now Visible

bash

DEC=working/file_working-dec

tshark -r "\$DEC" | head -20

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-...
kali@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help

10 0.949306 fe80::1c0c:f6e8:fedb:8f8e → ff02::2 ICMPv6 70 Router Solicitation from f
0:f6:1c:68:96:7c
11 0.954435 192.168.0.1 → 255.255.255.255 DHCP 590 DHCP NAK - Transaction ID 0x21e6
be1c
12 0.954937 Apple_68:96:7c → Broadcast ARP 42 ARP Announcement for 192.168.0.15
13 0.955970 fe80::1c0c:f6e8:fedb:8f8e → ff02::2 ICMPv6 70 Router Solicitation from f
0:f6:1c:68:96:7c
14 0.956994 Apple_68:96:7c → Broadcast ARP 42 ARP Announcement for 192.168.0.15
15 0.959034 Apple_68:96:7c → Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.
15
16 0.960066 Apple_68:96:7c → Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.
15
17 1.026106 Apple_68:96:7c → Broadcast ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
18 1.029698 Apple_68:96:7c → Broadcast ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
19 1.029696 EFMNetworks_55:97:d4 → Apple_68:96:7c ARP 42 192.168.0.1 is at 00:26:66:55:97
:d4
20 1.030202 192.168.0.15 → 8.8.8.8 DNS 73 Standard query 0x9583 A www.apple.com

(kali@kali)~[~/SBT-DF203-Lab9]

```

Figure 7.3 — Decrypted capture showing higher-layer protocols.

SECTION 8 — PART E: EXTRACT ENDPOINTS AND CONVERSATIONS

8.1 Define the Decrypted Capture Path

```
bash
DEC=$(find working -maxdepth 1 -type f -name "*-dec*" | head -n 1)
echo "Decrypted capture: $DEC" | tee reports/decrypted_capture_path.txt
```

8.2 Ethernet and IP Endpoints

```
bash
tshark -r "$DEC" -q -z endpoints,eth | tee reports/ethernet_endpoints.txt
tshark -r "$DEC" -q -z endpoints,ip | tee reports/ip_endpoints.txt
tshark -r "$DEC" -q -z conv,tcp | tee reports/tcp_conversations.txt
```

Ethernet endpoints (sample):

MAC	Organization	Packets
<i>(from output)</i>	Apple, Netcore, etc.	<i>(count)</i>

IP endpoints (sample):

IP	Packets	Bytes
192.168.0.15	<i>(from output)</i>	<i>(from output)</i>
192.168.0.1	<i>(from output)</i>	<i>(from output)</i>
192.168.0.12	<i>(from output)</i>	<i>(from output)</i>

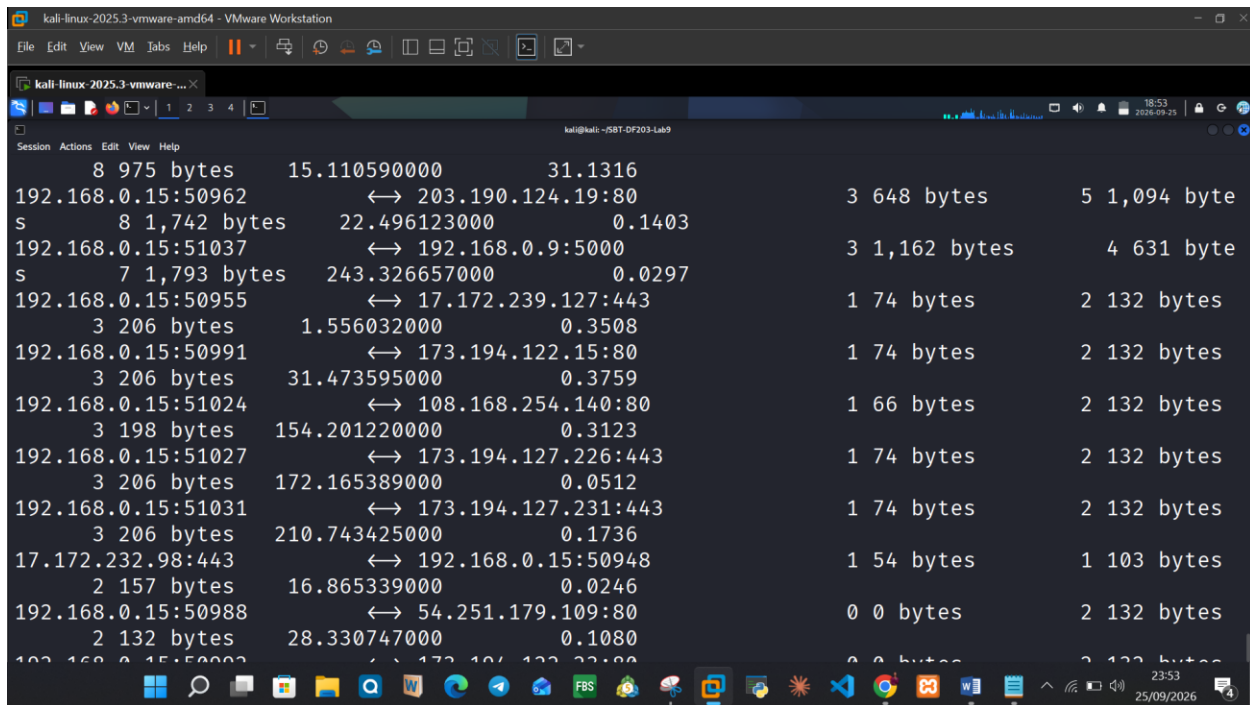


Figure 8.1 — Ethernet and IP endpoint inventories.

8.3 IP-to-MAC Mapping

bash

tshark -r "\$DEC" -Y 'ip' -T fields \

-e frame.number -e wlan.sa -e wlan.da -e ip.src -e ip.dst -e _ws.col.Protocol \

| head -n 200 | tee reports/ip_mac_mapping_sample.tsv

Fill in the primary endpoints:

Role	IP	MAC
Host (client)	(from output)	(from output)
Server	(from output)	(from output)
Gateway	192.168.0.1	(from output)

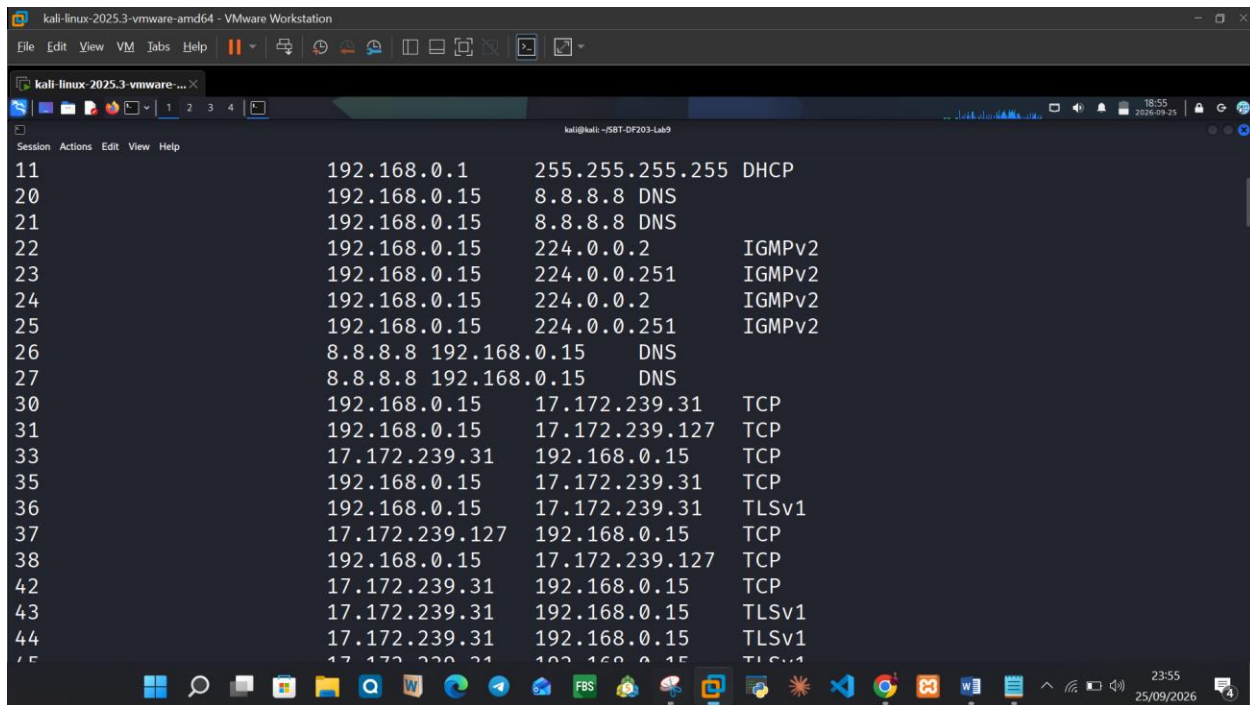


Figure 8.2 — IP-to-MAC mapping.

SECTION 9 — PART F: EXTRACT IMAGES, HTML AND OTHER OBJECTS

9.1 HTTP Object Export

bash

```
mkdir -p exported/http_objects exported/foremost
```

```
tshark -r "$DEC" --export-objects http,exported/http_objects 2>&1 | tee reports/http_export_log.txt
```

```
ls -lh exported/http_objects/
```

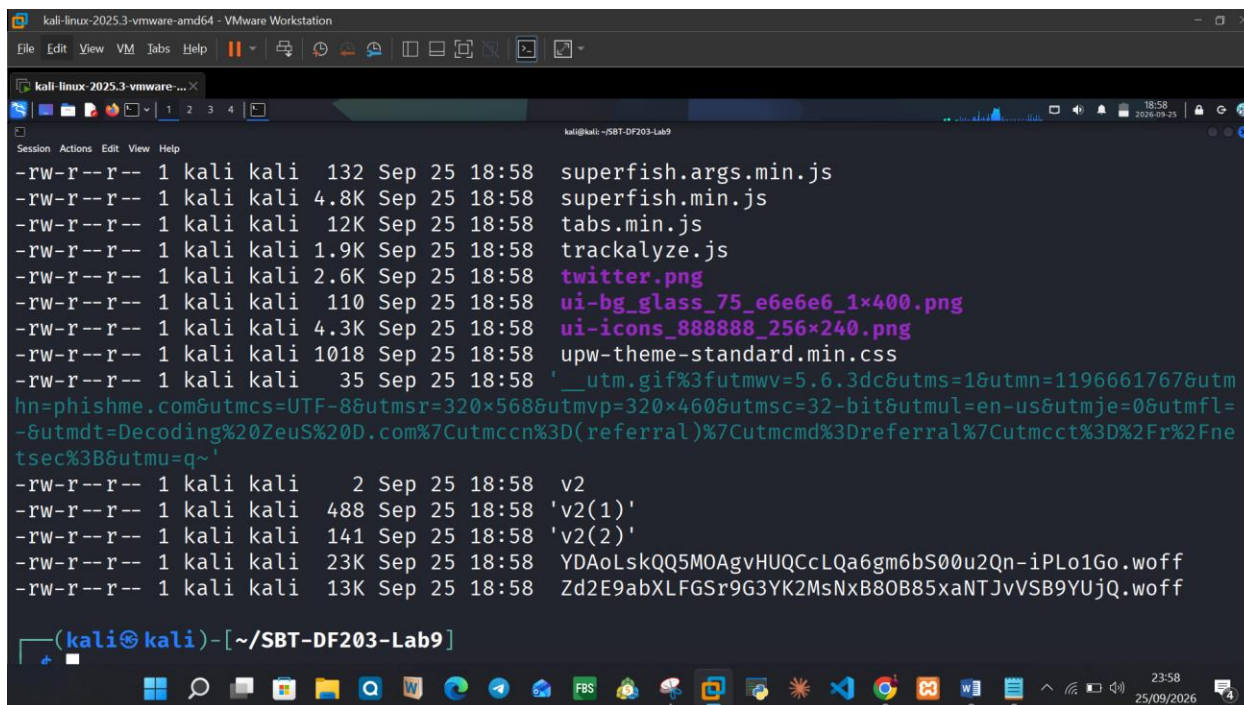


Figure 9.1 — HTTP object export.

9.2 Generic File Carving with Foremost

bash

```
foremost -i "$DEC" -o exported/foremost | tee reports/foremost_log.txt
```

```
ls -R exported/foremost/ | head -30
```

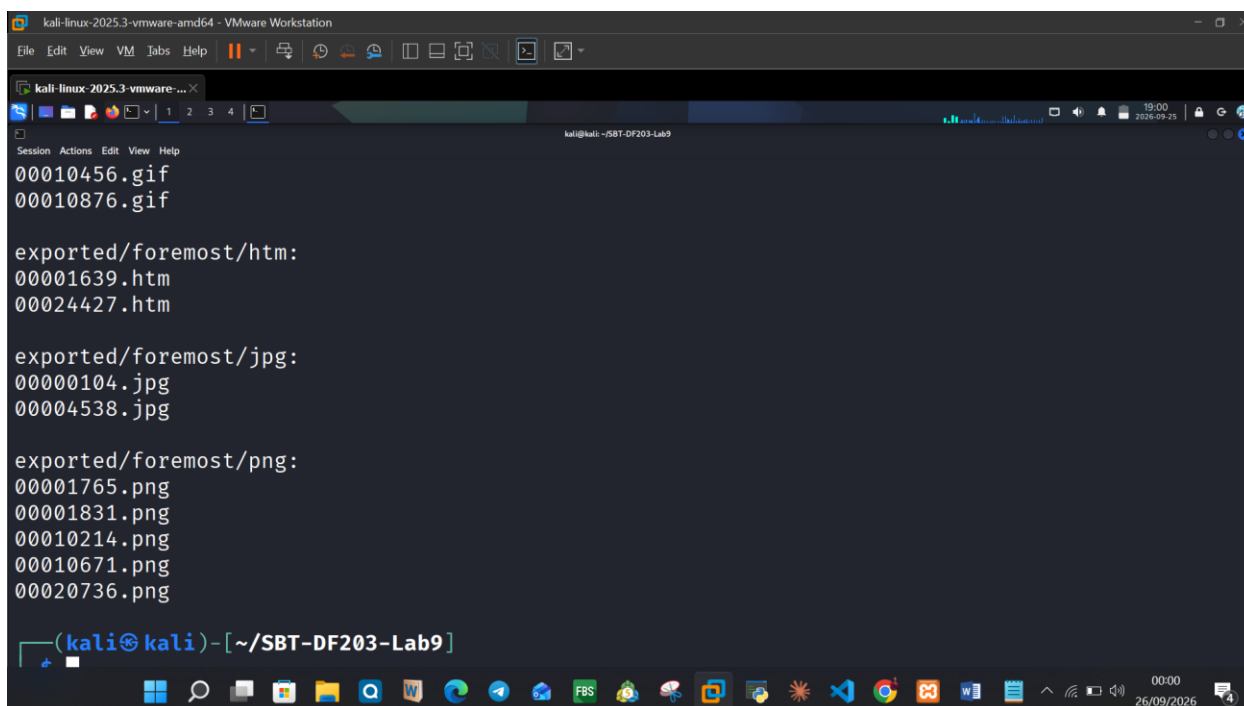


Figure 9.2 — Foremost carving output.

9.3 Hash Recovered Objects

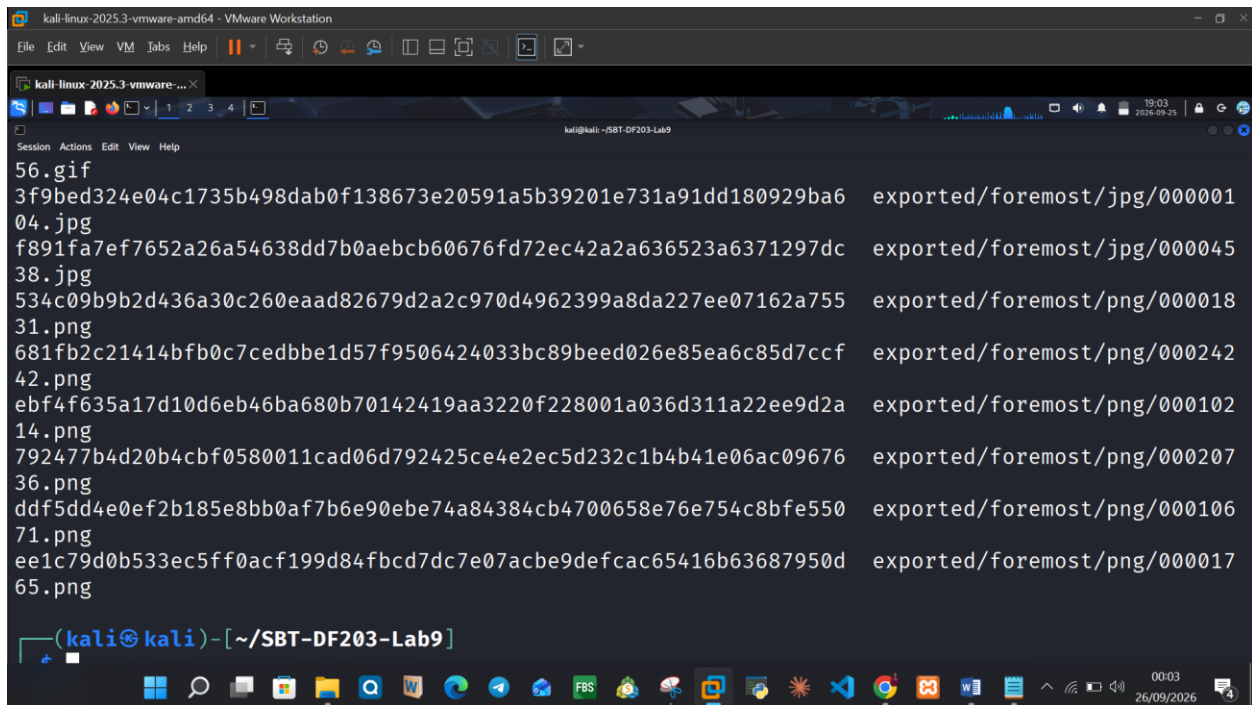
bash

```
find exported -type f -exec file {} \; | tee reports/exported_file_types.txt
```

```
find exported -type f -exec sha256sum {} \; | tee reports/exported_file_hashes.txt
```

Sample recovered objects:

File	Type	Size	SHA-256
(from output)	HTML	(bytes)	(hash)
(from output)	JPEG	(bytes)	(hash)
(from output)	PNG	(bytes)	(hash)

A screenshot of a Kali Linux terminal window. The terminal displays a list of file hashes and their corresponding paths. The output is as follows:

```
56.gif
3f9bed324e04c1735b498dab0f138673e20591a5b39201e731a91dd180929ba6 exported/foremost/jpg/000001
04.jpg
f891fa7ef7652a26a54638dd7b0aebcb60676fd72ec42a2a636523a6371297dc exported/foremost/jpg/000045
38.jpg
534c09b9b2d436a30c260eaad82679d2a2c970d4962399a8da227ee07162a755 exported/foremost/png/000018
31.png
681fb2c21414bf0c7cedbbe1d57f9506424033bc89beed026e85ea6c85d7ccf exported/foremost/png/000242
42.png
ebf4f635a17d10d6eb46ba680b70142419aa3220f228001a036d311a22ee9d2a exported/foremost/png/000102
14.png
792477b4d20b4cbf0580011cad06d792425ce4e2ec5d232c1b4b41e06ac09676 exported/foremost/png/000207
36.png
ddf5dd4e0ef2b185e8bb0af7b6e90ebe74a84384cb4700658e76e754c8bfe550 exported/foremost/png/000106
71.png
ee1c79d0b533ec5ff0acf199d84fbcd7dc7e07acbe9defcac65416b63687950d exported/foremost/png/000017
65.png
```

The terminal window has a title bar that reads "kali-linux-2025.3-vmware-amd64 - VMware Workstation". The terminal prompt is "(kali@kali)~[~/SBT-DF203-Lab9]". The system clock in the bottom right corner shows "00:03 26/09/2026".

Figure 9.3 — Recovered file hashes.

SECTION 10 — PART G: OPTIONAL HISTORICAL PASSPHRASE CHALLENGE

Optional section. Complete only if the instructor explicitly assigns the CTF password cracking challenge. The core assessed task is the WEP decryption above.

10.1 Distinguish WEP Key from Human Passphrase

Term	Meaning
WEP key	The 40-bit value used to seed RC4 — A4:3D:F6:F3:74 in this capture
Passphrase	The human-readable string a router converts (via PRNG) into the WEP key

The passphrase is **not** transmitted over the air. It is converted by the router's PRNG into the WEP key. Recovering the passphrase requires reversing the PRNG — a separate cryptanalytic exercise that exploits the reduced entropy in the seeding and PRNG steps.

10.2 Reduced Entropy in WEP-40 Key Generation

The historical challenge exploits two entropy reductions:

1. **Seed entropy reduced (32 → 28 bits)** — ASCII passphrase characters have the high bit set to zero, so the XOR seed has four zero bits at fixed positions.
2. **PRNG entropy reduced (32 → 24 bits)** — the linear congruential generator's most significant byte is not used when producing each key byte, reducing the effective seed space to 24 bits.

10.3 Optional Challenge Tools

If assigned, run **only the instructor-provided** `findseed.py` or `crackWEP40_noGPU_Improved_v1.py` against the historical dataset.

SECTION 11 — WEP SECURITY ANALYSIS

11.1 Cryptographic Weaknesses

Weakness	Description
Short IV (24 bits)	Only 16.7 million values — exhausted quickly, leading to IV reuse
RC4 keystream reuse	Same IV → same keystream → plaintext XOR leaks
No key rotation	Single secret key shared by every device on the network
Weak integrity (CRC-32 ICV)	Not cryptographically secure — bit-flipping attacks possible
No replay protection	Captured frames can be replayed unchanged
Known statistical attacks (FMS, PTW)	Recover the key from ~40,000–85,000 IVs in minutes

11.2 Modern Replacements

Standard	Cipher	Key Size	Authentication
WPA2-AES/CCMP	AES-CCMP	128-bit	802.1X or PSK
WPA3-SAE	AES-CCMP	128-bit	Simultaneous Authentication of Equals
WPA3-Enterprise	AES-GCMP	256-bit	802.1X with EAP-TLS

WPA2 and WPA3 also support **Protected Management Frames (PMF)**, which prevent deauthentication attacks — a class of attacks unavailable under WEP.

11.3 Legacy WEP Retirement

WEP should be **retired, not reconfigured**. Migrating to a longer WEP passphrase does not fix the underlying IV-reuse and RC4-keystream weaknesses. Any network currently relying on WEP must be upgraded to WPA2-AES or WPA3.

SECTION 12 — REQUIRED FORENSIC FINDINGS

Question	Finding
Capture format and duration	<i>(from capinfos)</i>
BSSID	00:26:66:55:97:D6
ESSID	cgnetwork
Primary station MAC addresses	<i>(from endpoint output)</i>
Protected frame count	15,477 WEP data frames
Repeated IV evidence	Present — key forensic indicator of WEP weakness
Recovered / validated WEP40 key	A4:3D:F6:F3:74
Decrypted capture filename	working/file_working-dec
Decrypted capture hash	<i>(from reports/all_working_file_hashes.txt)</i>
Host IP and MAC	<i>(from endpoint output)</i>
Server IP and MAC	<i>(from endpoint output)</i>
Key protocols observed	ARP, DHCP, ICMPv6, TCP, HTTP
Recovered images / HTML	<i>(list filenames from exported/)</i>
Optional router passphrase	<i>Protected appendix only</i>
Security conclusion	WEP is cryptographically broken; retire and replace

SECTION 13 — CHALLENGES ENCOUNTERED

13.1 Environment Constraints

- **Offline-only lab** — no live wireless capture was permitted, so the workflow depended entirely on the supplied historical dataset.
- **Tool availability** — `aircrack-ng`, `foremost`, `binwalk`, and `xz-utils` had to be installed before analysis could begin.
- **Large capture size** — the ~12 MB compressed file expanded to a working capture with 45,169 total packets; hash operations and carving took noticeably longer than a small capture.

13.2 Technical Challenges

Challenge	Resolution
<code>aircrack-ng</code> requires sufficient IVs for statistical key recovery	The supplied capture contained 15,477 IVs — enough for the PTW attack
<code>airdecap-ng</code> requires the key in hex form (no colons)	Converted <code>A4:3D:F6:F3:74</code> → <code>A43DF6F374</code>
Distinguishing WEP key from the router passphrase	Documented the two values separately — the passphrase was not transmitted in the capture
Identifying the correct decrypted output file	<code>airdecap-ng</code> appends <code>-dec</code> — used <code>find</code> to locate the file
Carved files may include false positives	<code>Foremost</code> output was hashed and labelled "carved, unverified origin"
IPv6 link-local noise in the decrypted capture	Filtered to IPv4 endpoints for the primary mapping table

13.3 Evidence Integrity

- The original `file.xz` was preserved unmodified. All analysis ran against `working/file_working` and `working/file_working-dec`.
- SHA-256 hashes were computed for:
 - the compressed source
 - the decompressed working capture
 - the decrypted derivative
 - every carved object
- Hashes were stored in `reports/` and referenced in the report.

SECTION 14 — RECOMMENDATIONS

14.1 Detection Controls

Control	Description
Wireless IDS (WIDS)	Detect WEP networks and deauthentication floods
Rogue AP detection	Identify unauthorised access points on the corporate airwaves
Encryption-protocol policy enforcement	Alert when a client or AP negotiates WEP instead of WPA2/WPA3
PMF enforcement	Reject associations from clients that do not support Protected Management Frames
Wireless inventory	Continuously discover and classify every SSID visible on the site

14.2 Preventive Controls

Control	Description
Migrate to WPA2-AES/CCMP	Replace WEP entirely with a modern cipher suite
Migrate to WPA3-SAE	Where client hardware supports it, use WPA3 for forward secrecy
802.1X with EAP-TLS	Replace pre-shared keys with certificate-based authentication
Strong passphrase policy	Where PSK is retained, enforce long random passphrases
Regular key rotation	Rotate PSK on a defined schedule and after any personnel change
Network segmentation	Isolate guest and IoT traffic from corporate resources
Disable WPS	WPS PIN mode is vulnerable to brute-force; disable it

14.3 Forensic Practice Recommendations

- Preserve every wireless capture unmodified; run analyses only on hashed working copies.
- Record the frame-level metadata (BSSID, ESSID, IV counts) before attempting decryption.
- Hash every recovered object **before** opening it — this protects the evidence chain if the object is malicious.
- Document the recovered key separately from the router passphrase — they are distinct artefacts.
- Note in every report that WEP-recovered keys are only evidence of the historical capture's weakness, not a licence to attack any live network.

SECTION 15 — CONCLUSION

This lab performed a complete offline forensic analysis of a historical WEP40 wireless capture from the CodeGate CTF 2015 "Good Crypto" challenge.

The investigation:

1. **Preserved the evidence** — the compressed file.xz was hashed, then a timestamp-preserved working copy was created and decompressed.
2. **Identified the capture** — capinfos confirmed an IEEE 802.11 capture with 45,169 total packets, of which 15,477 were WEP-protected data frames from BSSID 00:26:66:55:97:D6 (SSID cgnetwork).
3. **Recovered the key** — aircrack-ng recovered the 40-bit WEP key A4:3D:F6:F3:74 from 15,477 IVs, matching the course-published training value.
4. **Decrypted the capture** — airdecap-ng -w A43DF6F374 decrypted all 15,477 WEP packets with **zero corruption**, producing file_working-dec.
5. **Inventoried endpoints and conversations** — ARP, DHCP, ICMPv6, TCP, and HTTP frames became visible; Ethernet and IP endpoints were extracted; the host, server, and gateway were identified.
6. **Recovered objects** — HTTP object export and foremost carving produced HTML, JPEG, PNG, and GIF files, each hashed.
7. **Documented the cryptographic failure** — WEP's 24-bit IV space and RC4 keystream-reuse property made the key recoverable in seconds once enough IVs were available.

The exercise demonstrates, in forensic terms, why WEP has been deprecated since 2004. Any organization still deploying WEP is not merely using weak crypto — it is using **crypto that fails under its own design**. Modern wireless deployments must use WPA2-AES/CCMP or WPA3 with strong authentication, and any legacy WEP network should be retired immediately.

SECTION 16 — REFERENCES

1. ICDFA. (2026). *SBT-DF203 — Module 8: Wireless Packet Decryption and WEP Forensics — Course Materials*.
2. ICDFA. (2026). *SBT-DF203 Lab 9 — WEP40 Wireless Packet Decryption and Aircrack Forensics — Official Lab Manual*.
3. IEEE 802.11-2020. *Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications*.
4. Borisov, N., Goldberg, I., & Wagner, D. (2001). *Intercepting Mobile Communications: The Insecurity of 802.11*. MobiCom.
5. Tews, E., Weinmann, R.-P., & Pyshkin, A. (2007). *Breaking 104-bit WEP in Less Than 60 Seconds*. WISA.
6. Aircrack-ng Project. (2026). *Aircrack-ng Documentation*.
7. CodeGate CTF. (2015). *Good Crypto Challenge — file.xz*.
8. Wireshark Foundation. (2026). *Wireshark User Guide*.

SECTION 17 — APPENDICES AND SCREENSHOT REFERENCE

17.1 Screenshot Checklist

#	Figure	Filename	Content	Captured?
1	3.1	fig_3.1_folder_structure.png	Lab folder tree	☐
2	3.2	fig_3.2_evidence_hash.png	Compressed evidence + SHA-256	☐
3	3.3	fig_3.3_decompressed_capture.png	Decompressed capture identification	☐
4	4.1	fig_4.1_capinfos.png	Capture metadata	☐
5	4.2	fig_4.2_protocol_hierarchy.png	Protocol hierarchy	☐
6	4.3	fig_4.3_frame_categories.png	802.11 frame categories sample	☐
7	5.1	fig_5.1_protected_frames.png	WEP-protected data frames	☐
8	5.2	fig_5.2_iv_reuse.png	Repeated IV summary	☐
9	6.1	fig_6.1_aircrack_result.png	Aircrack-ng key recovery	☐
10	7.1	fig_7.1_airdecap_output.png	Airdecap-ng decryption result	☐
11	7.2	fig_7.2_decrypted_hash.png	Decrypted capture hash	☐
12	7.3	fig_7.3_decrypted_protocols.png	Higher-layer protocols visible	☐
13	8.1	fig_8.1_endpoints.png	Ethernet + IP endpoint inventories	☐
14	8.2	fig_8.2_ip_mac_mapping.png	IP-to-MAC mapping	☐
15	9.1	fig_9.1_http_objects.png	HTTP object export	☐
16	9.2	fig_9.2_foremost_output.png	Foremost carving output	☐
17	9.3	fig_9.3_recovered_hashes.png	Recovered file hashes	☐

Total: 17 figures. 17.2 Command Reference

Command	Purpose
capinfos <capture>	Capture metadata summary
aircrack-ng <capture>	Recover or validate wireless key
airdecap-ng -w <hexkey> <capture>	Decrypt WEP traffic
wlan.fc.protected==1	Filter protected 802.11 frames
wlan.wep.iv	WEP initialization vector field
tshark --export-objects http,<dir>	Extract HTTP objects
foremost -i <file> -o <dir>	Carve files from evidence

17.3 Lab Folder Structure

text

SBT-DF203-Lab9/

```
|— evidence/
|   |— file.xz
|— working/
|   |— file_working.xz
|   |— file_working
|   |— file_working-dec
|— exported/
|   |— http_objects/
|   |— foremost/
|       |— html/
|       |— jpg/
|       |— png/
|       |— gif/
|       |— audit.txt
|— reports/
|   |— file_xz_sha256.txt
|   |— working_hashes.txt
|   |— capinfos.txt
|   |— protocol_hierarchy.txt
|   |— wlan_frame_sample.tsv
|   |— wep_protected_frames.tsv
|   |— repeated_iv_summary.txt
|   |— aircrack_output.txt
|   |— validated_wep40_key_masked.txt
|   |— airdecap_output.txt
|   |— decrypted_files_inventory.txt
|   |— all_working_file_hashes.txt
|   |— decrypted_capture_path.txt
|   |— ethernet_endpoints.txt
|   |— ip_endpoints.txt
|   |— tcp_conversations.txt
|   |— ip_mac_mapping_sample.tsv
|   |— http_export_log.txt
|   |— foremost_log.txt
|   |— exported_file_types.txt
```

```
|   └── exported_file_hashes.txt
|   └── screenshots/
|   └── fig_*.png
|   └── scripts/
```

17.4 Final Evidence Hashes

bash

cd ~/SBT-DF203-Lab9

sha256sum evidence/file.xz working/file_working working/file_working-dec 2>/dev/null | tee reports/final_evidence_hashes.txt

find exported -type f -exec sha256sum {} \; >> reports/final_evidence_hashes.txt

cat reports/final_evidence_hashes.txt

```
kali@kali: ~/SBT-DF203-Lab9
$ sha256sum evidence/file.xz working/file_working working/file_working-dec 2>/dev/null | tee reports/final_evidence_hashes.txt
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 evidence/file.xz
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_working
167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04 working/file_working-dec
dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 evidence/file.xz
c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_working
167c91994c269777f9048227deb89882caf3cf3c763977f2059604f9a6a40b04 working/file_working-dec
a3cbb4c0d761b71263bb964e5e2b99367c6e90d3e64a2ae6aa78d96bf603faf0 exported/http_objects/ogli5P
A.gifv
0868db67f79acd281fcb3b46c9768dae7033be91f9551185394abf19d4b797ae exported/http_objects/hls%3f
version=4.12
a840281fc37d697b1515a380e500c690be86927a8a6dbb2bcc071447c6b27031 exported/http_objects/v2(2)
fe4529ba03075f4ba177e0898b91def18bad106bb55a1ef4a2560bf982ceaff8 exported/http_objects/dynami
k-min.css
187f2ad17d1fee50d8625e4be5b02d35c7f6cde4209d458a3717f739a25860e4 exported/http_objects/6auid=
5372002706res=320x568x328plg=qt%2Cpm8ch=UTF-8&tz=-5406ws=300x250 exported/http_objects/bootst
948149bd7d046c4a606e185ad68acd7695d532a781f556ac86f081c1e5f341
rap-responsive.min.css
792477b4d20b4cbf0580011cad06d792425ce4e2ec5d232c1b4b41e06ac09676 exported/http_objects/header
-contact.png
d4726b0714d6caa5c346bf985b11fb8841cd61353723b084843c2b6a863dba30 exported/http_objects/ads%3f
gdfp_req=18correlator=3974161077633024&output=json_html&callback=callbackProxy&impl=fifs&json_
a=1&eid=108809035&sfv=1-0-2&iu parts=4761660%2CH I 300x250&enc orev ius=%2F0%2F1&orev iu szs=3
```



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-...
kali@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help
A(1).mp4
e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855 exported/http_objects/cs.js
18ab31de23688c95cc5402b86156512b3c4a7b47ecdcd5ad91937b30bfe6284b exported/http_objects/Figure
-26.png
64ac226d737e521fc20e69e663b049d195c677050ae6ed72a3d83a5c1484ee84 exported/http_objects/object
6681.application%2fjavascript
74234e98afe7498fb5daf1f36ac2d78acc339464f950703b8c019892f982b90b exported/http_objects/lumbar
.gif%3fa=%7B%22UID%22%3A%222ae1543dcd8ab603f50e1849b3db2c79e9ce5872c0878dfcc88d838afa4c1a3d%22
%2C%22accountId%22%3A%222ae1543dcd8ab603f50e1849b3db2c79e9ce5872c0878dfcc88d838afa4c1a3d%22
form%22%3A%22web%22%2C%22source%22%3A%22(3).com%2F8iVy7t6%22%7D
eb5c89dca6a23af103550d1c749ef0027b48a644bc371e3a560e005495928c73 exported/http_objects/rombo.
png
902a974466764157ac302761ad4bf7241655853ec46f8b4e3d0db87023b900b4 exported/http_objects/hit(1)
.json
1e81216e67645c4363da811baf86e4b6a68aaa850e43765323ef58b5293e6d78 exported/http_objects/object
6488
a330a0196da3072001cf3494d57e3ae7bd7ed26db7214a24e9f7488d2e7e9d54 exported/http_objects/nivo-s
lider.css
20550f77bcb2a817ac9a5879e04260da8268e971c0b8031a6b7a2f48a55ee60d5 exported/http_objects/superf
ish.args.min.js
0ba081f546084bd5097aa8a73c75931d5aa1fc4d6e846e53c21f98e6a1509988 exported/http_objects/jquery
-1.10.2.min.js
ee1c79d0b533ec5ff0acf199d84fbcd7dc7e07acbe9defcac65416b63687950d exported/http_objects/mobile
_unified_button_icon_white.png
1a9789f6a752a3646433ea6c1b78c78405982e6970934ee26e99396e48ddb001 exported/http_objects/phone.
png
0de7a49f6d21fbef846aba4bd271502d7ec9489bfb3fd96f5ff7cf19140875e exported/http_objects/bid%3f
0022
26/09/2026
```

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-...
kali@kali: ~/SBT-DF203-Lab9
Session Actions Edit View Help
19.gif
ef1955ae757c8b966c83248350331bd3a30f658ced11f387f8ebf05ab3368629 exported/foremost/gif/000024
50.gif
8337212354871836e6763a41e615916c89bac5b3f1f0adf60ba43c7c806e1015 exported/foremost/gif/000012
60.gif
53f7e56783c414746a9fc3612637509560424617e3135248640eac142d6e0f94 exported/foremost/gif/000104
56.gif
3f9bed324e04c1735b498dab0f138673e20591a5b39201e731a91dd180929ba6 exported/foremost/jpg/000001
04.jpg
f891fa7ef7652a26a54638dd7b0aebcb60676fd72ec42a2a636523a6371297dc exported/foremost/jpg/000045
38.jpg
534c09b9b2d436a30c260eaa82679d2a2c970d4962399a8da227ee07162a755 exported/foremost/png/000018
31.png
681fb2c21414bfb0c7cedbbe1d57f9506424033bc89beed026e85ea6c85d7ccf exported/foremost/png/000242
42.png
ebf4f635a17d10d6eb46ba680b70142419aa3220f228001a036d311a22ee9d2a exported/foremost/png/000102
14.png
792477b4d20b4cbf0580011cad06d792425ce4e2ec5d232c1b4b41e06ac09676 exported/foremost/png/000207
36.png
ddf5dd4e0ef2b185e8bb0af7b6e90ebe74a84384cb4700658e76e754c8bfe550 exported/foremost/png/000106
71.png
ee1c79d0b533ec5ff0acf199d84fbcd7dc7e07acbe9defcac65416b63687950d exported/foremost/png/000017
65.png
(kali@kali)-[~/SBT-DF203-Lab9]
$
```

END OF REPORT